
NASH BARGAINING PREFERENCE OPTIMIZATION

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Paper under double-blind review

ABSTRACT

Aligning one large language model (LLM) to several objectives requires two decisions: how to represent preferences within each objective and how to choose one compromise across objectives. The first decision is difficult because pairwise judgments can be cyclic in realistic settings. For a medical-advice prompt, annotators may prefer a concise answer to a verbose one, the verbose answer to an incomplete one, yet the incomplete answer to the concise one because it includes a critical safety caveat. No scalar ranking can satisfy all three comparisons. Nash Learning from Human Feedback avoids this restriction by treating pairwise preferences as game payoffs, but its multi-objective extensions still require an aggregation rule. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective through a regularized preference game and applies Nash bargaining as an aggregation rule to its improvement over a common reference policy. Our main technical contribution is a practical optimizer for these policy-dependent game values: a constrained KL-proximal update handles zero surplus at the reference, closed-form opponent reweighting exposes weak responses, and pairwise log-ratio regression removes the intractable policy normalizer while learning directly from binary comparisons. Under exact population updates, the final iterate achieves convergence with rate $O(1/T)$. Experiments compare preference-based and reward-model-based multi-objective baselines under an independent evaluator on three objective panels. No method leads every panel, and we report that plainly; what NBPO does attain is the property the bargaining axioms actually assert — it stays above the reference on every objective of every panel, and it is the only method that does so without either fitting a scalar reward per objective or being given a floor to respect — together with numerically exact invariance to rescaling of the objective coordinates, which the fixed-weight and max-min alternatives do not have. A matched ablation isolates the aggregation rule from the size of the proximal step, and an audit of the evaluation prompts is reported in full.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a standard approach for aligning large language models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard RLHF learns a scalar reward from pairwise comparisons. Direct Preference Optimization removes the separately trained reward model, but still assumes that comparisons are generated by an implicit scalar reward through a Bradley–Terry (BT) model (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches therefore require every response to admit one scalar score whose ordering explains all pairwise preferences.

This assumption can fail even within a single alignment objective. Human judgments often reflect several latent attributes whose relative importance changes with the prompt and the pair being compared. Consider three answers to a medical-advice question. Response A is concise and actionable but omits an important caveat; response B is comprehensive but verbose; and response C is less complete but includes the critical safety warning. Annotators may prefer A to B for relevance, B to C for completeness, and C to A for safety and reliability. These comparisons are cyclic: $A \succ B$, $B \succ C$, and $C \succ A$. Fitting the first two comparisons with a scalar reward requires $r(A) > r(B) > r(C)$, which contradicts the third. As a result, optimizing the fitted scalar reward can favor a response that appears best under the learned ranking but is still beaten by another response according to the original pairwise judgments.

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a second problem: after representing each objective without imposing transitivity, which compromise should one deployed policy realize? Scalar multi-objective methods expose this trade-off through weights, priorities, constraints, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). Recent game-theoretic methods preserve general pairwise feedback but combine objectives through Blackwell target sets or a worst-objective criterion (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026).

Classical Nash bargaining offers a priority-free way to select one compromise. It maximizes the product of each objective’s improvement over a common fallback (Nash, 1950). In alignment, the reference policy provides a natural fallback: an aligned policy should improve each objective relative to the model that would otherwise be deployed. The classical Nash solution is Pareto optimal and invariant to independent changes in the units used for different objectives. Prior alignment work has applied Nash welfare to learned scalar rewards, while game-based preference methods have focused mainly on minimax or Blackwell solutions (Zhong et al., 2024a; Navon et al., 2022; Zhang et al., 2026). What remains is a practical method for Nash bargaining when every objective value is itself defined by a policy-dependent preference game rather than by a fixed scalar reward.

We introduce Nash Bargaining Preference Optimization (NBPO) for this setting. Within each objective, NBPO evaluates the learner against an adaptive opponent that emphasizes responses exposing the learner’s weaknesses, while a KL penalty keeps the opponent near the reference policy. Across objectives, it maximizes the Nash product of the resulting improvements over that same reference. The standard bargaining properties are useful in LLM alignment because they ensure that no objective falls below the fallback at the population optimum, no other feasible policy can improve one objective without worsening another, and arbitrary judge-specific units do not alter the selected compromise.

The main technical challenge is optimization, not re-deriving the classical bargaining axioms. Three issues arise simultaneously: the reference policy has zero improvement and lies on the boundary of the logarithmic objective; each adaptive opponent depends on the policy being optimized; and the ideal exponential policy update contains an intractable normalizing constant. We address these issues with a constrained KL-proximal formulation, closed-form opponent reweighting, a low-dimensional dual update whose weights are inversely proportional to achieved improvements, and partition-free pairwise log-ratio regression from binary comparisons.

Our contributions are:

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, thereby separating within-objective preference representation from across-objective compromise selection.
- We derive a practical training procedure that handles the zero-surplus reference boundary, alternates closed-form opponent updates with policy fitting, and learns the policy update without fitting scalar rewards or evaluating a language-model partition function.
- We establish $O(1/T)$ last-iterate convergence for exact population proximal updates and evaluate the method against matched preference-optimization controls and reward-model-based multi-objective baselines under independent held-out metrics.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y \mid x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z \mid x) \in [0, 1]$ satisfying

$$P_k(y \succ z \mid x) + P_k(z \succ y \mid x) = 1, \quad P_k(y \succ y \mid x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z \mid x) = P_k(y \succ z \mid x) - \frac{1}{2}, \quad \Delta_k(y, z \mid x) = -\Delta_k(z, y \mid x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot \mid x), z \sim \nu(\cdot \mid x)} \Delta_k(y, z \mid x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex, and downward-closed feasible set with at least one strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G}: s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective’s pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x))\}. \quad (6)$$

The comparator minimizes the learner’s pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z \mid x) := \mathbb{E}_{y \sim \pi(\cdot \mid x)} \Delta_k(y, z \mid x) \quad (7)$$

be the learner’s expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k, \pi}^*(z \mid x) = \frac{\pi_{\text{ref}}(z \mid x) \exp\{-m_k^\pi(z \mid x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot \mid x)} \exp\{-m_k^\pi(z' \mid x)/\beta_k\}}, \quad (8)$$

$$V_{k, \beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot \mid x)} \exp\{-m_k^\pi(z \mid x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k, β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_{\pi} V_{k, \beta_k}(\pi)]_{x, y} = \mathcal{D}(x) Q_k^\pi(y \mid x), \quad Q_k^\pi(y \mid x) := \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot \mid x)} \Delta_k(y, z \mid x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^\pi(y \mid x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y \mid x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

The assumption states that at least one policy improves all objectives over the fallback. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_\beta(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_\beta := \{v \in \mathbb{R}^K : d \leq v \leq V_\beta(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_\beta$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_\beta, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. In LLM alignment, they mean that the population solution improves every objective over the reference, cannot be Pareto dominated by another feasible policy, and does not change merely because different judges report their values in different numerical units. Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective and its gradient are undefined there. Second, each objective's opponent depends on the policy being optimized. Third, the ideal exponential policy update contains a language-model normalizer that cannot be computed. This section addresses the three issues with a constrained proximal step, alternating opponent and policy updates, and pairwise log-ratio regression.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_\pi F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^\pi(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors in (15) emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves the constrained KL-proximal problem

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. Assumption 1 supplies a feasible positive-surplus policy, while π_{ref} serves only as the initial proximal center.

The dual of (16) is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (17)$$

For fixed λ , denote the unique inner maximizer by

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

Danskin's theorem yields the low-dimensional dual gradient

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the same inverse-surplus weighting visible in (15): objectives with smaller achieved improvements receive larger weights, without choosing those weights in advance.

5.2 SAMPLED POLICY FITTING AND DUAL UPDATE

Fix outer stage t and a dual vector λ . Because the opponent $\nu_{k,\pi}^*$ depends on the learner, we approximately solve (18) by alternating two operations. At inner iteration r , first construct $\nu_{k,\pi(r)}^*$ for each objective. Then hold these opponents fixed and fit the next policy. Define

$$Q_k^{(r)}(y | x) := \mathbb{E}_{z \sim \nu_{k,\pi(r)}^*(\cdot | x)} \Delta_k(y, z | x).$$

The ideal fixed-opponent update is

$$\pi^{(r+1)}(y | x) \propto \pi_t(y | x) \exp \left\{ \eta_t \sum_{k=1}^K \lambda_k Q_k^{(r)}(y | x) \right\}. \quad (21)$$

Its normalizer disappears when comparing two responses. Let

$$h_t(\pi; x, y, y') := \log \frac{\pi(y | x)}{\pi_t(y | x)} - \log \frac{\pi(y' | x)}{\pi_t(y' | x)}. \quad (22)$$

Then (21) implies

$$h_t(\pi^{(r+1)}; x, y, y') = \eta_t \sum_{k=1}^K \lambda_k \left(Q_k^{(r)}(y | x) - Q_k^{(r)}(y' | x) \right). \quad (23)$$

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , preference models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T , dual steps M , policy-opponent steps R .

- 1: Estimate $d_k = V_{k, \beta_k}(\pi_{\text{ref}})$ and set $\pi_0 \leftarrow \pi_{\text{ref}}$.
 - 2: **for** $t = 0, \dots, T - 1$ **do**
 - 3: Initialize positive weights $\lambda^{(0)}$ (warm-start from the previous stage when available).
 - 4: **for** $m = 0, \dots, M - 1$ **do**
 - 5: $\hat{\pi}^{(m)} \leftarrow \text{POLICYFIT}(\pi_t, \lambda^{(m)}, R)$ using opponent reweighting and (26).
 - 6: Estimate $\hat{s}(\hat{\pi}^{(m)})$ and update $\lambda^{(m+1)}$ by (27).
 - 7: **end for**
 - 8: $\tilde{\pi}_{t+1} \leftarrow \text{POLICYFIT}(\pi_t, \lambda^{(M)}, R)$.
 - 9: Set $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$ if all held-out empirical surpluses are positive; otherwise retain π_t .
 - 10: **end for**
 - 11: **return** π_T .
-

To estimate the right-hand side, sample (y, y') from a full-support pair proposal $\xi(\cdot, \cdot | x)$ and sample $z_k \sim \nu_{k, \pi^{(r)}}^*(\cdot | x)$. Query independent binary judgments

$$B_k \sim \text{Bernoulli}(P_k(y \succ z_k | x)), \quad B'_k \sim \text{Bernoulli}(P_k(y' \succ z_k | x)),$$

and set

$$Z_k^{(r)} := B_k - B'_k. \quad (24)$$

Because

$$\mathbb{E}[Z_k^{(r)} | x, y, y'] = Q_k^{(r)}(y | x) - Q_k^{(r)}(y' | x), \quad (25)$$

the weighted binary label is an unbiased target for the desired log-ratio. We therefore fit

$$\mathcal{L}_{\text{reg}}^{(t, \lambda, r)}(\pi) = \mathbb{E} \left[h_t(\pi; x, y, y') - \eta_t \sum_{k=1}^K \lambda_k Z_k^{(r)} \right]^2, \quad (26)$$

where the expectation follows the sampling process above. Recomputing the opponents after each policy fit produces a fixed point that satisfies the optimality condition of (18); Appendix A.5 gives the population argument.

After approximately solving the weighted policy problem, we estimate its surplus vector and update the positive dual weights by

$$\lambda^{(m+1)} = \Pi_{\Lambda} \left[\lambda^{(m)} - \gamma_m \left(\hat{s}(\pi_t^*(\lambda^{(m)})) - (\lambda^{(m)})^{-1} \right) \right], \quad (27)$$

where the inverse is elementwise and Π_{Λ} keeps the numerical weights in a positive bounded box. The indices have distinct roles: r alternates opponent construction and policy fitting for fixed λ ; m updates the dual weights within one proximal stage; and t commits the next outer policy.

The held-out check is an implementation safeguard rather than a finite-sample guarantee. The convergence result below concerns exact population solutions of (16).

5.3 LAST-ITERATE CONVERGENCE

Because Algorithm 1 returns the final outer policy rather than an average of past policies, we analyze the last population iterate.

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (28)$$

A constant proximal step therefore gives an $O(1/T)$ final-iterate Nash-welfare gap. Appendix A.6 gives the proof.

6 EXPERIMENTS

Setup. We evaluate fixed objective panels on ULTRAFEEDBACK, TL;DR, and SAFERLHF. The primary backbone and reference are Llama-3.1-8B-Instruct. A Qwen3-8B replication of the ULTRAFEEDBACK panel is reported on general-capability benchmarks only: its response pool was built on prompts that include the panel’s own evaluation prompts (Appendix E), so an objective-wise win rate there would be measured in sample, whereas the benchmark prompts are external to every training set in this paper. A prompted Qwen3-32B model supplies objective-specific training comparisons, queried in both presentation orders. Held-out evaluation uses models that supply none of the training labels. Appendix C reports data splits, generation, judging, optimization, and compute; Appendix J gives the objective rubrics and judge prompts.

Baselines and evaluation. We include two baseline families. Preference-based baselines are PROSPER/MaxEntBW, equal-weight utilitarian and maxmin controls over the same game-valued objectives, Uniform-INPO, MOPO, and single-objective HT-MNPO. Reward-model-based baselines are MaxMin-RLHF, Rewarded Soups, and Panacea (Chakraborty et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). For methods that produce a family of policies, one operating point must be selected using validation data only. Appendix F records the reward models, policy optimizer, selection rule, and compute required for these runs.

Our primary cross-method metric is independent objective-wise win rate against the common reference,

$$W_k^{\text{eval}}(\pi) := \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x), y_{\text{ref}} \sim \pi_{\text{ref}}(\cdot|x)} P_k^{\text{eval}}(y \succ y_{\text{ref}} | x). \quad (29)$$

All methods are ranked by the same held-out evaluators, so this metric does not privilege the NBPO training objective. Evaluation prompts are held out by prompt-text hash rather than by assumption: Appendix E reports the audit of every arm’s training pairs against its panel’s evaluation prompts, the one panel on which the split had failed, and the rebuild that repairs it. Table 2 reports the worst and average W_k^{eval} across each panel, with every objective listed in Appendix B. The adaptive game-value surpluses s_k remain useful for diagnosing whether an implementation realizes the NBPO population target, but we report them only as a secondary method-aligned diagnostic in the appendix. Table 1 evaluates general capability.

Main multi-objective comparison. Table 2 reports the primary comparison. Every entry is measured on prompt-disjoint held-out prompts (698 on ULTRAFEEDBACK, 400 on the other two panels), judged by Llama-3.3-70B-Instruct in both presentation orders, with two policy samples against two reference samples per prompt and objective. The stated uncertainty is a prompt-level bootstrap standard error of the reported statistic, resampling prompts once per replicate and recomputing the minimum and the mean across objectives jointly; because the arms share prompts, the paired differences quoted below are far better resolved than these marginal errors suggest. Appendix B reports the per-objective rates behind each entry.

The comparison does not support a single ordering, and we do not claim one. No method attains the highest worst-objective win rate on all three panels: Panacea leads it on ULTRAFEEDBACK and TL;DR, while the game-aggregation controls lead it on SAFERLHF. NBPO is second of eleven on TL;DR, sixth on ULTRAFEEDBACK, and ninth on SAFERLHF, where it is the panel’s compression rather than its own collapse that puts it there: apart from Uniform-INPO and the single-objective corners, every arm on that panel lands within 0.085 of every other.

What the measurements do support is the weaker property the bargaining axioms actually assert, and Figure 1 shows it directly. Individual rationality on every coordinate — staying above the reference on every objective of every panel — holds for four of the eleven methods: NBPO, MOPO, Rewarded Soups and Panacea. The other three reach it by construction rather than as a consequence: Rewarded Soups and Panacea interpolate or condition on a fitted scalar reward per objective and their operating point is chosen on validation data by the largest minimum utility, and MOPO designates a primary objective and imposes hand-set lower bounds on the rest. NBPO is the only one that neither fits a scalar reward per objective nor is given a floor to respect; individual rationality falls out of the bargaining solution, which is what the axioms predict. The seven methods that do not attain it fail badly rather than marginally: Uniform-INPO reaches 0.540 on ULTRAFEEDBACK but 0.228 on TL;DR, MaxMin-RLHF’s alternation ends its last round at 0.346 on TL;DR’s conciseness, and the utilitarian and absolute max-min controls fall to 0.415 and 0.365 there. NBPO’s worst and

Table 1: General-capability evaluation after multi-objective training. The first four columns are held-out win rates against the corresponding reference policy; IFEval and TruthfulQA use their official metrics. Higher is better.

method						
<i>Backbone and reference policy: Llama-3.1-8B-Instruct</i>						
Base = $\pi_{\text{ref}} = \pi_0$	0.500	0.500	0.500	0.500	0.762	0.541
NBPO (ours)	0.517	0.525	0.524	0.506	0.756	0.542
PROSPER/MaxEntBW	0.515	0.546	0.541	0.560	0.630	0.543
Game-utilitarian	0.522	0.544	0.548	0.516	0.617	0.544
Game-absolute-maxmin	0.532	0.536	0.548	0.525	0.638	0.544
Game-surplus-maxmin	0.505	0.532	0.524	0.503	0.636	0.548
Uniform-INPO	0.527	0.517	0.535	0.503	0.645	0.541
MOPO	0.508	0.512	0.516	0.513	0.749	0.542
HT-MNPO (honesty)	0.549	0.528	0.522	0.519	0.717	0.544
HT-MNPO (truthfulness)	0.536	0.531	0.517	0.525	0.675	0.547
HT-MNPO (helpfulness)	0.530	0.550	0.546	0.494	0.649	0.542
HT-MNPO (instr. following)	0.515	0.529	0.533	0.541	0.739	0.541
MaxMin-RLHF	0.534	0.522	0.506	0.547	0.723	0.546
Rewarded Soups	0.534	0.540	0.515	0.497	0.741	0.544
Panacea	0.579	0.540	0.518	0.506	0.739	0.551
<i>Backbone and reference policy: Qwen3-8B, reasoning trace suppressed</i>						
Base = $\pi_{\text{ref}} = \pi_0$	0.500	0.500	0.500	0.500	0.815	0.544
NBPO (ours)	0.512	0.562	0.534	0.529	0.810	0.546
PROSPER/MaxEntBW	0.505	0.535	0.519	0.497	0.808	0.544
Game-utilitarian	0.509	0.528	0.511	0.497	0.810	0.545
Game-absolute-maxmin	0.510	0.531	0.514	0.503	0.806	0.545
Game-surplus-maxmin	0.500	0.534	0.523	0.516	0.804	0.544
Uniform-INPO	0.491	0.498	0.496	0.481	0.821	0.545
MOPO	0.498	0.499	0.497	0.468	0.815	0.543
HT-MNPO (honesty)	0.493	0.500	0.503	0.513	0.815	0.544
HT-MNPO (truthfulness)	0.503	0.517	0.508	0.528	0.813	0.544
HT-MNPO (helpfulness)	0.491	0.497	0.498	0.519	0.811	0.545
HT-MNPO (instr. following)	0.491	0.491	0.490	0.509	0.819	0.544
MaxMin-RLHF	0.500	0.457	0.491	0.532	0.824	0.544
Rewarded Soups	0.499	0.475	0.484	0.550	0.813	0.544
Panacea	0.478	0.469	0.485	0.519	0.826	0.543

Table 2: Primary cross-method evaluation under independent objective-wise win rates against the common reference. We report the worst and average objective win rate in each panel, with prompt-level bootstrap standard errors. Higher is better. Every arm here is trained on prompts disjoint from the evaluation prompts of its panel; the ULTRAFEDBACK NBPO entry is the rebuild described in Appendix E, because the originally released arm was not.

method	ULTRAFEDBACK		TL;DR		SAFERLHF	
	worst W^{eval}	avg. W^{eval}	worst W^{eval}	avg. W^{eval}	worst W^{eval}	avg. W^{eval}
Base = π_{ref}	0.500	0.500	0.500	0.500	0.500	0.500
NBPO (ours)	0.524 $\pm$ 0.007	0.535 $\pm$ 0.006	0.528 $\pm$ 0.011	0.558 $\pm$ 0.007	0.696 $\pm$ 0.013	0.718 $\pm$ 0.012
PROSPER/MaxEntBW	0.496 $\pm$ 0.008	0.541 $\pm$ 0.007	0.444 $\pm$ 0.015	0.592 $\pm$ 0.007	0.760 $\pm$ 0.013	0.777 $\pm$ 0.012
Game-utilitarian	0.467 $\pm$ 0.008	0.500 $\pm$ 0.007	0.415 $\pm$ 0.016	0.595 $\pm$ 0.007	0.780 $\pm$ 0.012	0.787$\pm$0.012
Game-absolute-maxmin	0.459 $\pm$ 0.008	0.499 $\pm$ 0.008	0.365 $\pm$ 0.014	0.557 $\pm$ 0.006	0.781$\pm$0.012	0.782 $\pm$ 0.012
Game-surplus-maxmin	0.502 $\pm$ 0.009	0.549 $\pm$ 0.007	0.491 $\pm$ 0.013	0.602 $\pm$ 0.007	0.778 $\pm$ 0.012	0.780 $\pm$ 0.012
Uniform-INPO	0.540 $\pm$ 0.009	0.580$\pm$0.007	0.228 $\pm$ 0.015	0.519 $\pm$ 0.007	0.447 $\pm$ 0.016	0.471 $\pm$ 0.012
MOPO	0.507 $\pm$ 0.007	0.511 $\pm$ 0.007	0.513 $\pm$ 0.012	0.546 $\pm$ 0.007	0.720 $\pm$ 0.013	0.730 $\pm$ 0.012
HT-MNPO, best corner	0.528 $\pm$ 0.008	0.547 $\pm$ 0.007	0.502 $\pm$ 0.011	0.517 $\pm$ 0.008	0.470 $\pm$ 0.010	0.542 $\pm$ 0.010
MaxMin-RLHF	0.537 $\pm$ 0.007	0.557 $\pm$ 0.006	0.346 $\pm$ 0.014	0.564 $\pm$ 0.007	0.705 $\pm$ 0.014	0.736 $\pm$ 0.012
Rewarded Soups	0.534 $\pm$ 0.008	0.557 $\pm$ 0.007	0.506 $\pm$ 0.014	0.569 $\pm$ 0.007	0.728 $\pm$ 0.013	0.742 $\pm$ 0.011
Panacea	0.554$\pm$0.007	0.572 $\pm$ 0.007	0.560$\pm$0.012	0.602$\pm$0.007	0.703 $\pm$ 0.013	0.726 $\pm$ 0.011

average objective rates also differ by the least across the table, though only just: at most 0.030 against MOPO’s 0.033, and MOPO’s spread is small mainly because it stays close to the reference. None of this is the stronger claim that Nash aggregation maximises the minimum in finite samples, which our own matched ablation does not establish.

General capability. Table 1 asks what each method costs outside its own objective panel. On the Llama-3.1-8B-Instruct backbone the pairwise columns separate the methods little, but IFEval separates them sharply, and it separates them against the game-aggregation controls rather

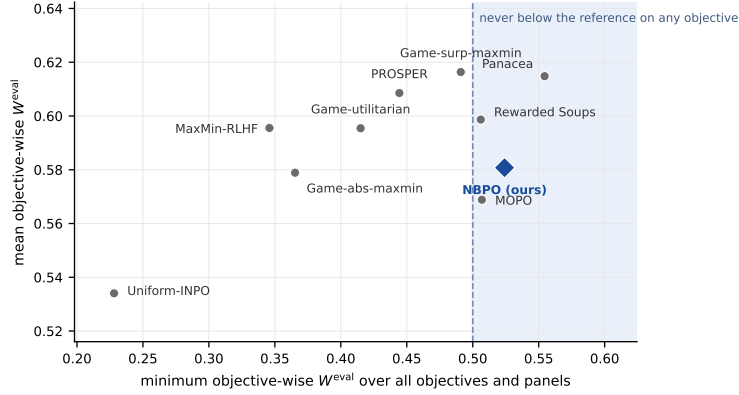


Figure 1: Each method’s weakest objective against its average, pooled over every objective of all three panels. The horizontal axis is the lowest objective-wise W^{eval} the method attains anywhere in Table 2; the vertical axis is its mean over the same objectives. Points to the right of the dashed line never fall below the reference on any objective of any panel, which is the property the bargaining axioms assert. Height above a point’s own horizontal position is average performance bought by sacrificing the weakest coordinate.

than against NBPO: the reference policy scores 0.762, NBPO 0.756, and every control that shares its response pool and budget falls to between 0.617 and 0.645. Of the fifteen trained arms NBPO gives up the least instruction-following capability, 0.006, and MOPO is next at 0.013; the utilitarian and max-min aggregations of the same game values give up more than twenty times as much. On the Qwen3-8B backbone NBPO has the highest score in three of the four pairwise columns and on TruthfulQA, and the three reward-model baselines land at or below the reference on Arena-Hard and below it on AlpacaEval. These prompts come from Arena-Hard, AlpacaEval, MT-Bench, IFEval and TruthfulQA and are external to every training set in this paper, so this block is the one place the Qwen3-8B panel can be read without the caveat of Appendix E.

Separating representation and aggregation. Table 2 compares each method at its own selected configuration, so a gap there can come from the aggregation rule or from how far that rule happens to move the policy. The rules do not move it equally: on the frozen pools the weight vectors they induce differ in ℓ_1 norm by factors of 14 to 55, because the Nash dual carries raw $\lambda_k = 1/s_k$ with $\|\lambda\|_1 = \sum_k 1/s_k$ while the utilitarian rule sits at $\lambda = 1$ and the max-min rules on the simplex. Table 3 therefore rebuilds the ablation under a matched design. On each panel the five variants share the response pool, the dual solution’s η_t , the opponent fixed point, the target estimator of (25), the optimiser budget, and the total ℓ_1 weight norm — the controls are rescaled to the Nash arm’s, since Nash’s own magnitude is fixed by the dual — so they differ only in the direction of the aggregation weights or in whether the objective value is the adaptive game or a fixed reference. Every entry sits closer to the reference than its counterpart in Table 2 because these fits use the η_t that the pooled-entropy rule selects and a smaller learning rate, not because of the weight normalisation; levels are therefore not comparable across the two tables, and the comparison within a column is what this table is for.

Two readings change. Nash aggregation survives matching in direction but not in magnitude: it has the highest matched worst-objective rate on ULTRAFEEDBACK and is within 0.006 of the best on the other two panels, and it is never behind any rule by more than the paired interval. Its advantage over the two max-min rules is resolved only on SAFERLHF, at +0.023 [0.012, 0.034] over surplus max-min and +0.026 [0.015, 0.038] over absolute max-min; against the utilitarian rule the paired differences are +0.006, +0.005 and +0.000, all straddling zero. The much larger unmatched gaps therefore reflect step size, not the bargaining weights. The representation comparison changes sign and loses significance: with the step matched, the adaptive game and the fixed reference are indistinguishable on all three panels, at +0.005 [−0.007, 0.016], −0.006 [−0.018, 0.014] and −0.001 [−0.011, 0.008] on the worst objective. We previously read an unmatched 0.036 advantage for the fixed reference on ULTRAFEEDBACK as evidence against the adaptive opponent; that reading does not survive the

Table 3: Matched component ablation on the held-out prompts of all three panels. Within a panel the five variants share the frozen response pool, η_t , the opponent fixed point, the target estimator, the optimiser budget and the total ℓ_1 weight norm, and differ only in the objective value or in the direction of the aggregation weights. Higher is better; the bold entry is the best in its column.

objective value	aggregation	ULTRAFEEDBACK		TL;DR		SAFERLHF	
		worst	avg.	worst	avg.	worst	avg.
adaptive game	Nash (NBPO)	0.509$\pm$0.007	0.513 $\pm$ 0.006	0.500 $\pm$ 0.010	0.508 $\pm$ 0.007	0.631 $\pm$ 0.011	0.633 $\pm$ 0.010
adaptive game	utilitarian	0.503 $\pm$ 0.007	0.507 $\pm$ 0.007	0.495 $\pm$ 0.010	0.500 $\pm$ 0.007	0.631 $\pm$ 0.011	0.633 $\pm$ 0.011
adaptive game	surplus maxmin	0.506 $\pm$ 0.007	0.509 $\pm$ 0.006	0.488 $\pm$ 0.011	0.504 $\pm$ 0.007	0.608 $\pm$ 0.011	0.610 $\pm$ 0.010
adaptive game	absolute maxmin	0.506 $\pm$ 0.007	0.515$\pm$0.007	0.503 $\pm$ 0.010	0.510$\pm$0.007	0.605 $\pm$ 0.011	0.606 $\pm$ 0.010
fixed reference	Nash	0.503 $\pm$ 0.007	0.510 $\pm$ 0.007	0.506$\pm$0.009	0.508 $\pm$ 0.007	0.632$\pm$0.011	0.635$\pm$0.011

Table 4: Pairwise update-score sign flips (%) under three independent objective rescalings on a frozen ULTRAFEEDBACK pool. Lower is better; zero is exact invariance.

selection rule	$c^{(A)}$	$c^{(B)}$	$c^{(C)}$	mean
NBPO (ours)	0.01	0.01	0.00	0.01
Utilitarian	5.66	6.31	5.85	5.94
Uniform-INPO	5.66	6.31	5.85	5.94
MOPO	5.66	6.31	5.85	5.94
PROSPER/MaxEntBW	7.35	6.63	6.71	6.90
Surplus maxmin	16.02	13.57	0.00	9.86
Absolute maxmin	16.32	0.00	16.32	10.88

matched design, and we report the honest conclusion that this experiment does not separate the two representations. Appendix I isolates where the finite-pool realisation departs from the population target.

Scale invariance and optimization diagnostics. Because affine invariance is a standard Nash-bargaining property rather than a new alignment theorem, Table 4 checks whether the implemented update preserves it numerically. For each frozen response pair, we compare the sign of the aggregation-induced policy-update score before and after independently rescaling the objective coordinates. NBPO changes this direction for only 0.01% of pairs; alternative fixed-weight or maxmin rules change more often. Appendix H reports temperature, comparator-sampling, fixed-point, dual, and regression diagnostics needed to identify the remaining gap between the population objective and the neural implementation.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026). NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Reward-based multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize weakest-objective criteria, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These methods are important empirical baselines because they explicitly fit or use scalar objective rewards, whereas NBPO learns from general pairwise feedback without assuming that each objective is transitive.

Social choice and bargaining. The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2024a); Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

NBPO separates two decisions in multi-objective preference alignment: representing possibly cyclic comparisons within each objective and selecting one compromise across objectives. It uses regularized preference games for the first decision and the classical Nash bargaining rule for the second. The bargaining guarantees are inherited once the policy-induced game-value set satisfies the standard conditions; our main contribution is the constrained and partition-free optimization procedure that makes this construction trainable from binary comparisons. The exact population proximal iteration achieves an $O(1/T)$ last-iterate Nash-welfare gap.

Several limitations remain, and the completed comparison sharpens them. The matched component ablation does not establish an empirical benefit from adaptive game values: once the proximal step is matched, the adaptive game and the fixed reference are statistically indistinguishable on all three panels, so this experiment neither credits nor debits the adaptive opponent. The matched ranking among aggregation rules is also weaker than the unmatched one: Nash has the best or near-best matched worst-objective rate on every panel and is never behind by more than the paired interval, but its advantage over the max-min rules is resolved only on SAFERLHF and its advantage over the utilitarian rule is not resolved anywhere. The cross-method comparison likewise does not show NBPO attaining the highest worst-objective win rate: Panacea leads it on two panels and the game-aggregation controls lead it on SAFERLHF. What the measurements do support is the weaker property the bargaining axioms actually assert: NBPO stays above the reference on every objective of every panel, and it is the only method that does so without either fitting a scalar reward per objective or being handed a floor to respect — the three others that manage it (MOPO, Rewarded Soups, Panacea) do one or the other by construction. The neural algorithm also approximates the policy-opponent fixed point, dual solution, and regression target from finite samples, whereas the convergence theorem assumes exact population updates, and Appendix D shows what that costs: a second outer stage on SAFERLHF, run through the released stage runner and accepted by the algorithm’s own gate, returns $+0.003$ on the gated surplus and -0.008 on the reported win rate — the per-stage approximation error is at least as large as the progress the extra stage makes. The joint-improvement assumption can fail when no policy improves every objective over the chosen reference; in that case a positive-surplus Nash solution does not exist. Improvements in game value should not be interpreted as per-prompt safety guarantees.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 specifies the constrained training procedure. Appendix A provides the proximal-dual derivation and population convergence proof; Appendices C, H and J give the experimental protocol for both configurations, the optimization and regression diagnostics, and the judge details; Appendix E gives the evaluation-prompt contamination audit and the rebuild it prompted. Every policy behind a number in Tables 1, 2 and 3 is released as a public checkpoint, together with the frozen response pools, the dual solver artifacts (weights, surpluses and fixed-point residuals), the pair files with their target columns, and the judge prompts and per-prompt verdicts behind every win rate; the repository is withheld here only to preserve anonymity and is included in the submission package. Seeds, exact sampling budgets, and the prompt-hash lists used for the disjointness audit are part of that release.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin's theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (30)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_\nu J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k, β_k} is continuous, $V_{\beta}(\pi)$ is bounded. The definition of $\mathcal{G}_{\beta}$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_{\beta}$, choosing policies π_n with $d \leq v_n \leq V_{\beta}(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_{\beta}$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_{\beta}(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_{\beta}(\pi_1) + (1 - \alpha)V_{\beta}(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_{\beta}$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_{\beta}$ with $v > d$, some policy π satisfies $v \leq V_{\beta}(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k, \beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_{\beta}(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_{\beta}, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatewise monotonicity implies Pareto optimality. Strict concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

The positive guarantees in Table 5 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus

Table 5: Classical population guarantees of common aggregation rules under joint improvement. The Nash row follows from the standard bargaining solution; the other rows are supported by the finite counterexamples below.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maxmin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maxmin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

For the uniform reference,

$$\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$$

and hence

$$d = (-1/3, -1/12).$$

First, joint improvement is feasible. The policy

$$\bar{\pi} = (1/4, 1/4, 1/2)$$

is only a feasibility witness; it gives

$$\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$$

Therefore

$$V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$$

In contrast, solving the equal-weight utilitarian problem $\max_{\pi} V_1(\pi) + V_2(\pi)$ gives

$$\pi_U = (0, 1/5, 4/5).$$

For this policy,

$$\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$$

so

$$V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$$

Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

Absolute-maxmin failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again,

$$d = (-1/3, -1/12).$$

Joint improvement is feasible, as witnessed by

$$\bar{\pi} = (1/3, 7/15, 1/5),$$

for which

$$V(\bar{\pi}) = (-4/15, -1/60)$$

and therefore

$$s(\bar{\pi}) = (1/15, 1/15) > 0.$$

Absolute maxmin instead solves

$$\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$$

whose linear-program solution is

$$\pi_{AM} = (0, 4/5, 1/5).$$

For this policy,

$$\pi_{AM}^T \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{AM}^T \Delta_2 = (-1/10, -1/10, 2/5),$$

so

$$V(\pi_{AM}) = (-1/10, -1/10)$$

and

$$s(\pi_{AM}) = (7/30, -1/60).$$

Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed 1/6.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at 1/6. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION AND REGRESSION FIXED POINT

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin's theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Regression fixed point. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (31)$$

At a fixed point of the alternating inner solve, the opponents in (26) are exactly $\nu_{k, \pi_t^*(\lambda)}^*$. Equation (25) therefore makes the conditional mean of the binary target equal to the right-hand side of (31). The squared-loss conditional-mean property implies that a population minimizer of (26) satisfies this pairwise identity. Full support of the pair proposal identifies all log-probability differences; normalization then recovers $\pi_t^*(\lambda)$.

A.6 PROOF OF THEOREM 1

Let $h(\pi) = \mathbb{E}_{x \sim \mathcal{D}} \sum_y \pi(y | x) \log \pi(y | x)$, so its Bregman divergence is $D(\pi \| \pi_t)$. Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (32)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (32) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (33)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (35)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (36)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (35), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_{\text{ref}}). \quad (37)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (28).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \|\pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (38)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because $\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (39)$$

Applying this inequality to $s = s(\pi_T)$ and using (28) proves (38). $\square$

B OBJECTIVE-WISE EVALUATION AND GAME-SURPLUS DIAGNOSTICS

Table 6 is the objective-wise counterpart of the primary external comparison in Table 2; all entries use the same independent evaluator and the same prompt split as that table. Table 7 retains the currently available adaptive game-value surpluses only as a diagnostic of the NBPO-aligned objective; because this metric is defined by NBPO’s game construction, it should not be the sole headline ranking of unrelated algorithms.

Table 7 gives the objective-wise decomposition of the existing game-surplus results. Its NBPO row comes from an earlier campaign that fitted the policy with the pipeline’s RONPO-style regression rather than the sampled pair target of (24) and (26). That arm is a different objective, not a differently tuned NBPO: it optimises a RONPO-style regression rather than the finite-pool target of (25) and (26), so no setting of η_t reproduces it. It is therefore kept only as the historical diagnostic it is, and every NBPO row in the main text is a matched finite-pool fit. These values are useful for diagnosing which objectives each policy sacrifices, but they are defined by the adaptive game evaluation introduced by NBPO and are therefore secondary to the common external metric in Table 6.

C IMPLEMENTATION AND HYPERPARAMETERS

Following prior preference-optimization and multi-objective alignment work, we report response generation, preference scoring, policy optimization, method-specific loop parameters, and compute separately (Munos et al., 2024; Zhang et al., 2025; Zhong et al., 2024b; Zhang et al., 2026). Tables 8 and 9 record both configurations that appear in the paper, because they differ. The arms of the primary comparison sit at $\eta_t = 1$ with one opponent reweighting per dual update. That is not the value the pooled-entropy rule of Table 14 selects, and the discrepancy is why we measured both: on every panel, at both η_t values, we recorded whether the fitted policy beats $\pi = \pi_t$ on a prompt-disjoint held-out split of its own pair set. Table 16 reports the result, and it selects $\eta_t = 1$ on all three panels by a wide margin. The matched-ablation arms deliberately use the pooled-entropy η_t , the iterated fixed point, and a correspondingly smaller learning rate, which is a factor of 15 to 20 smaller in the proximal target the regression is asked to realize; that is why their levels are not comparable to the primary table’s. A cell is left empty only where the corresponding quantity does not apply.

Implementation audit. The entries above distinguish the released configuration from the rebuilt one. The validation split is prompt-level, and we audited it rather than assuming it: every arm’s training pairs were hashed against its panel’s evaluation prompts. Section E reports what that audit found, including one panel on which it failed. The opponent fixed point is iterated rather than approximated: the released arms recorded $R = 1$, one opponent reweighting per dual update, and every rebuilt arm uses $R = 8$ with damping 0.9 and a recorded residual below 10^{-9} (Table 15); a sensitivity check at the selected η_t found $R \in \{1, 8, 64\}$ to give the same solution to solver tolerance,

Table 6: Objective-wise held-out win rates W_k^{eval} against the common reference, on prompt-disjoint evaluation sets judged by Llama-3.3-70B-Instruct.

method	obj. 1	obj. 2	obj. 3	obj. 4	average
<i>ULTRAFEEDBACK: instruction following, truthfulness, honesty, helpfulness</i>					
Base = π_{ref}	0.500	0.500	0.500	0.500	0.500
NBPO	0.528	0.524	0.539	0.547	0.535
PROSPER/MaxEntBW	0.518	0.496	0.544	0.605	0.541
Game-utilitarian	0.479	0.467	0.504	0.550	0.500
Game-absolute-maxmin	0.472	0.459	0.501	0.565	0.499
Game-surplus-maxmin	0.502	0.517	0.574	0.603	0.549
Uniform-INPO	0.540	0.556	0.608	0.617	0.580
MOPO	0.507	0.508	0.512	0.516	0.511
MaxMin-RLHF	0.554	0.537	0.567	0.569	0.557
Rewarded Soups	0.559	0.534	0.553	0.582	0.557
Panacea	0.568	0.554	0.570	0.594	0.572
<i>TL;DR: coverage, faithfulness, conciseness, helpfulness</i>					
Base = π_{ref}	0.500	0.500	0.500	0.500	0.500
NBPO	0.572	0.593	0.528	0.539	0.558
PROSPER/MaxEntBW	0.673	0.590	0.444	0.661	0.592
Game-utilitarian	0.692	0.599	0.415	0.675	0.595
Game-absolute-maxmin	0.694	0.493	0.365	0.675	0.557
Game-surplus-maxmin	0.578	0.727	0.613	0.491	0.602
Uniform-INPO	0.399	0.668	0.780	0.228	0.519
MOPO	0.574	0.557	0.513	0.540	0.546
MaxMin-RLHF	0.710	0.517	0.346	0.682	0.564
Rewarded Soups	0.571	0.618	0.580	0.506	0.569
Panacea	0.649	0.616	0.584	0.560	0.602
<i>SAFERLHF: helpfulness, harmlessness</i>					
Base = π_{ref}	0.500	0.500	–	–	0.500
NBPO	0.740	0.696	–	–	0.718
PROSPER/MaxEntBW	0.794	0.760	–	–	0.777
Game-utilitarian	0.794	0.780	–	–	0.787
Game-absolute-maxmin	0.781	0.784	–	–	0.782
Game-surplus-maxmin	0.778	0.781	–	–	0.780
Uniform-INPO	0.447	0.495	–	–	0.471
MOPO	0.740	0.720	–	–	0.730
MaxMin-RLHF	0.768	0.705	–	–	0.736
Rewarded Soups	0.755	0.728	–	–	0.742
Panacea	0.750	0.703	–	–	0.726

so the 0.9999 residual reported for the released $\eta_t = 1$ arm is a symptom of that arm’s degenerate inner solution rather than of R itself. The query budgets are also separated into the two accounting rows above rather than mixed. One gap remains open and is stated as such: the Qwen panel’s released arm was solved at $\eta_t = 1$ with $R = 1$ and the (24) draw, and was not rebuilt.

D A SECOND OUTER STAGE

Every arm in this paper is $T = 1$: the dual is solved once on a pool generated by π_{ref} , and one proximal step is taken. The convergence result of Theorem 1 is about T , so the untested knob is the outer loop itself. We ran one genuine second stage on SAFERLHF, through the released stage runner rather than by hand, so that its own preconditions are enforced.

The setup is the algorithm’s, not a variant of it. π_1 is the reported SAFERLHF arm. Its 500 training prompts were split off from the panel’s 600 so that the monitoring set the Algorithm 1 gate uses is disjoint from the training pairs as well as from the 400 final-evaluation prompts; the runner refuses to start otherwise, and the audited intersections are all zero. The reference policy does not change between stages, so its response samples are reused and only π_1 ’s are decoded afresh. The training preference matrix was rejudged in full, 44,000 cells in both presentation orders, the dual was re-solved on the new pool ($\lambda = (3.256, 3.558)$), projected KKT residual 0, $R = 8$), the pair targets rebuilt, and the policy refit for the same 300 updates at the same η_t and learning rate.

The gate accepted the candidate: its held-out surpluses are $(0.240, 0.148)$, positive on both objectives, so Nash welfare is defined and π_2 was promoted. What the stage bought is another matter. Measuring π_1 on the same monitoring prompts with the same judge — without which the accepted value means nothing — gives a minimum surplus of 0.145 against π_2 ’s 0.148. On the panel’s 400 evaluation

Table 7: Secondary adaptive game-value surplus diagnostic for the existing single-run checkpoints. This NBPO-aligned metric is retained for mechanism analysis rather than used as the primary cross-method ranking. It is computed on each panel’s frozen response pool, so it is an in-sample quantity for every panel by construction; that is why it is a diagnostic and not a ranking, and it is also why the Qwen3-8B block appears here but not in Table 6 (Appendix E).

method	obj. 1	obj. 2	obj. 3	obj. 4	min <i>s</i>	avg <i>s</i>
ULTRAFEEDBACK on Llama-3.1-8B-Instruct: instruction following, truthfulness, honesty, helpfulness						
NBPO (ours)	0.039	0.049	0.055	0.109	0.039	0.063
PROSPER/MaxEntBW	−0.035	−0.051	−0.046	0.068	−0.051	−0.016
Utilitarian	−0.064	−0.077	−0.070	0.027	−0.077	−0.046
Absolute maxmin	−0.043	−0.062	−0.051	0.059	−0.062	−0.024
Surplus maxmin	−0.014	−0.000	0.020	0.108	−0.014	0.028
Uniform-INPO	−0.008	0.022	0.020	0.079	−0.008	0.028
MOPO	0.031	0.026	0.022	0.032	0.022	0.028
HT-MNPO (helpfulness)	−0.057	−0.073	−0.067	0.050	−0.073	−0.037
HT-MNPO (truthfulness)	−0.050	−0.049	−0.041	0.023	−0.050	−0.029
HT-MNPO (honesty)	0.022	0.027	0.025	0.075	0.022	0.037
HT-MNPO (instr. following)	0.008	0.026	0.032	0.065	0.008	0.033
ULTRAFEEDBACK on Qwen3-8B: helpfulness, truthfulness, honesty, instruction following						
NBPO (ours)	0.072	0.018	0.043	0.031	0.018	0.041
PROSPER/MaxEntBW	0.039	−0.002	0.016	0.017	−0.002	0.018
Utilitarian	0.038	0.012	0.016	0.012	0.012	0.019
Absolute maxmin	0.041	0.005	0.018	0.012	0.005	0.019
Surplus maxmin	0.038	0.004	0.014	0.009	0.004	0.016
Uniform-INPO	−0.005	−0.001	−0.001	−0.003	−0.005	−0.002
MOPO	−0.002	−0.001	−0.005	0.001	−0.005	−0.001
HT-MNPO (helpfulness)	−0.002	−0.001	−0.001	0.002	−0.002	−0.001
HT-MNPO (truthfulness)	0.017	0.001	0.016	0.014	0.001	0.012
HT-MNPO (honesty)	0.000	0.005	0.001	−0.003	−0.003	0.001
HT-MNPO (instruction following)	−0.004	0.004	0.001	0.002	−0.004	0.001
TL;DR on Llama-3.1-8B-Instruct: coverage, faithfulness, conciseness, helpfulness						
NBPO (ours)	0.198	0.079	−0.001	0.178	−0.001	0.114
PROSPER/MaxEntBW	0.205	0.031	−0.068	0.175	−0.068	0.086
Utilitarian	0.221	0.072	−0.046	0.188	−0.046	0.109
Absolute maxmin	0.213	0.005	−0.079	0.182	−0.079	0.080
Surplus maxmin	0.081	0.090	0.018	0.066	0.018	0.064
Uniform-INPO	−0.051	−0.052	−0.011	−0.050	−0.052	−0.041
MOPO	0.061	0.009	−0.001	0.052	−0.001	0.030
HT-MNPO (coverage)	0.051	−0.010	−0.007	0.045	−0.010	0.020
HT-MNPO (faithfulness)	0.074	−0.020	−0.026	0.060	−0.026	0.022
HT-MNPO (conciseness)	0.126	−0.037	−0.079	0.106	−0.079	0.029
HT-MNPO (helpfulness)	0.037	−0.025	−0.035	0.048	−0.035	0.006
SAFERLHF on Llama-3.1-8B-Instruct: helpfulness, harmlessness						
NBPO (ours)	0.118	0.101	—	—	0.101	0.110
PROSPER/MaxEntBW	0.151	0.116	—	—	0.116	0.134
Utilitarian	0.157	0.138	—	—	0.138	0.148
Absolute maxmin	0.154	0.144	—	—	0.144	0.149
Surplus maxmin	0.150	0.143	—	—	0.143	0.147
Uniform-INPO	−0.077	−0.055	—	—	−0.077	−0.066
MOPO	0.102	0.096	—	—	0.096	0.099
HT-MNPO (helpfulness)	−0.024	−0.049	—	—	−0.049	−0.036
HT-MNPO (harmlessness)	−0.023	−0.050	—	—	−0.050	−0.037

prompts, judged by the same independent evaluator as every other arm in this paper, π_2 reaches worst-objective 0.688 and average 0.716, against 0.696 and 0.718 for π_1 .

So a second outer stage costs a full pipeline pass — a 44,000-cell rejudging, a re-solve and a retrain — and returns +0.003 on the quantity the algorithm gates on and −0.008 on the metric the paper reports, both inside the noise of their respective sample sizes. We read this as the finite-sample approximation at each stage dominating the $O(1/T)$ improvement the exact population iteration would deliver: the theorem assumes exact updates, and at $T = 2$ the error introduced by regenerating and rejudging the pool is at least as large as the progress the extra stage makes. That is a limitation of the implementation rather than of the bargaining formulation, but it is the limitation that matters for anyone deciding whether to pay for a second stage, and on this evidence they should not.

E EVALUATION-PROMPT CONTAMINATION AUDIT

Every arm’s training pairs were hashed against its own panel’s evaluation prompts, rather than relying on the split having been taken correctly. Table 10 reports the result. Two panels and most arms are clean, and two failures are worth stating plainly.

On ULTRAFEEDBACK the released NBPO arm was fitted on pairs whose prompts include 629 of the panel’s 698 evaluation prompts. The cause is bookkeeping rather than method: this panel’s response pool was regenerated for the present campaign and its prompts were drawn independently of the earlier campaign from which the evaluation list was inherited, so the two collided. The released stage runner refuses to start when the training, monitoring and evaluation prompt sets are not pairwise disjoint, and this arm was built by calling the dual solver and pair builder directly rather than through it, which is how the collision went unnoticed; the rebuild is checked against the same condition explicitly. The reward-model baselines escaped it because their pair builder takes the evaluation pool as an explicit exclusion list, and the fixed-reference and adaptive-game arms of the earlier campaign escaped it because they predate the regenerated pool. The consequence is that the released arm was the only ULTRAFEEDBACK entry measured partly in sample, and the bias runs in its own favour, so the panel’s conclusion — that NBPO does not lead the worst-objective rate there — is conservative under it. We nonetheless rebuilt the arm on the identical configuration with the 629 offending prompts and their pairs removed (37,800 $\rightarrow$ 34,026 training rows over 5,671 prompts), and Table 2 reports the rebuilt arm. The rebuild lands on worst-objective 0.524 and average 0.535, against 0.524 and 0.540 for the arm it replaces: the defect was real and had to be repaired, but on this panel it was not what produced the number. The rebuilt arm also fits its own target better, which is the reason to prefer it beyond cleanliness — held-out normalized MSE 0.983 against 0.998, and held-out R^2 +0.017 against +0.002 (Table 16).

The matched ablation arms inherit a much smaller share of the same collision, 51 of 698, because they are fitted on a subsample of that pool. All five of them share one pair set by construction, so whatever bias those 51 prompts introduce is common to every column of Table 3 and cancels in the paired differences that table reports; the comparison it is built for is between arms on the same prompts, not between an arm and the reference.

On the ULTRAFEEDBACK/Qwen panel the same collision is total: the 300-prompt response pool contains all 100 evaluation prompts, and 99 of them survive into every arm’s training pairs. That panel is therefore reported only through the general-capability benchmarks of Table 1, whose prompts come from Arena-Hard, AlpacaEval, MT-Bench, IFEval and TruthfulQA and are external to every training set here. We do not report an objective-wise win rate for the Qwen backbone, and the reader should treat it as a backbone-transfer check on general capability rather than as a fourth panel of the primary comparison.

F REWARD-MODEL BASELINE PROTOCOL

Reward-model baselines require additional choices that can materially affect the comparison, so Table 11 records every one of them for the rows these methods occupy in Tables 1 and 2. All three share the preference arms’ response pool, initialisation and update budget, so the comparison isolates how the objective-wise rewards are combined.

G ADDITIONAL EXTERNAL EVALUATION

The general-capability table reserves IFEval and TruthfulQA for their official task-specific scoring. Because safety uses a different metric and direction, HarmBench is reported separately in Table 12 rather than mixed into Table 1.

H OPTIMIZATION DIAGNOSTICS AND ROBUSTNESS

Table 13 separates the numerical dual solve from the harder neural-realization problem, and the split is lopsided: the dual is easy and the realization is not. At the selected step size the dual converges by $M = 4,000$ (dual residual 5×10^{-7} , against 1.3×10^{-2} at $M = 400$; a tenfold larger budget

buys 1.8×10^{-15} and the same surplus vector), and the opponent budget R changes nothing at all — $R = 1$, $R = 8$ and $R = 64$ return the same multipliers and the same surplus vector, differing only in how sharply the fixed point is certified. The failure of the fixed point that this paper reports at $R = 1$ belongs to $\eta_t = 1$, where $\eta_t \sum_k \lambda_k = 25$ collapses the inner solution onto a single response; it is a symptom of the step size, not of the opponent budget.

The one genuinely sensitive quantity in the dual is the opponent temperature, and its effect has a sign that should not be read as quality: weakening the opponent raises the surplus by construction. As β grows the opponent’s entropy rises from 0.963 to $\ln 4 = 1.386$, and the minimum surplus rises monotonically with it, reaching its maximum exactly at the fixed-reference limit where the opponent is uniform. A surplus measured against one opponent therefore cannot be compared with a surplus measured against another, which is why the representation comparison in Appendix I is settled on the external win rate instead.

Everything expensive is on the other side of the split. The dual is solved to a KKT residual of 10^{-15} ; the regression that turns that solution into a policy (Table 16) reaches a held-out R^2 between $+0.017$ and $+0.098$. The remaining error lives there, not in the dual, and the gap between the two is nine orders of magnitude.

I DIAGNOSING THE ADAPTIVE-GAME GAP

An earlier version of this paper reported that the simpler fixed-reference Nash variant had the better point estimate, and asked which of three explanations was responsible: the adaptive population objective may not help on this pool, the neural regression may fail to realize its target, or the pool may contain too little disagreement between objectives for an adaptive opponent to matter. All three now have measurements. The first thing they show is that the comparison itself was confounded, and that once it is unconfounded the advantage disappears in both directions: the matched rows of Table 3 put the two representations within 0.006 of each other on all three panels, with every paired interval straddling zero.

The comparison was not matched. Fixed reference is the same solver at $\beta \rightarrow \infty$, which collapses the opponent onto π_{ref} and makes the opponent fixed point of (21) hold by construction: its solver artifact records a fixed-point residual of exactly 0. The adaptive arm it was compared against was solved with a single policy–opponent step, whose measured residual was 0.9999. The published contrast was therefore between a converged arm and an unconverged one, and attributed the difference to the representation. The two surpluses are not comparable either: each is measured against its own opponent, and a weaker opponent yields a larger surplus by definition. Every arm in Table 3 is now solved to the same fixed point (Table 15) and takes the same total ℓ_1 proximal step, and the decisive comparison is the external win rate W^{eval} , which does not depend on either side’s opponent.

The regression did fail to realize its target, and the fix was the data. Table 16 measures the fit directly. The released ULTRAFEEDBACK policy’s log-ratio change was uncorrelated with the target it was fitted to: normalized MSE 1.0005 on its own training pairs, $R^2 = -0.0005$, and a training loss flat from step 10 to step 300. A policy that does not realize its target is not an approximation of the NBPO update at all, which is the first thing that had to be fixed before any comparison between weight rules could mean anything.

Refitting the same configuration on the contamination-free pair set of Appendix E changes that, and so does simply measuring the other two panels, which had never been diagnosed at their released η_t . At the selected step size every panel now beats the zero predictor on prompt-disjoint held-out pairs: normalized MSE 0.983, 0.902 and 0.959, with held-out R^2 of $+0.017$, $+0.098$ and $+0.041$ and held-out correlations between 0.14 and 0.33. The finite-pool target is therefore realizable and does generalize, modestly, at this data scale — the earlier reading that it does not was drawn from one contaminated fit and from rebuilds at a much smaller η_t . Those small- η_t rebuilds are in the table too, and they are the failure case: at the η_t the pooled-entropy rule selects, the same optimizer overshoots a target fifteen to twenty times smaller and lands at held-out normalized MSE 2.65 and 3.50. This is why the primary comparison selects η_t by the held-out realization criterion rather than by pooled entropy; the two rules disagree, both are computed without touching evaluation data, and only one

of them is about whether the policy realizes the update. A control that precomputes μ twice and differences the two passes gives $\|h\|_{\text{rms}} = 0$ exactly, so none of these numbers is measurement noise.

The η_t rule was set on training data and the evaluation agrees with it. Selecting η_t by the held-out realization criterion rather than by pooled entropy is a decision made entirely on the training side, and it is worth reporting what the metric it never saw says about it. We retrained all three panels at the η_t the pooled-entropy rule selects, changing nothing else, and judged them with the same independent evaluator:

panel	η_t	held-out nMSE (selection criterion)	worst W^{eval} (not used to select)
ULTRAFEEDBACK	1.0 (selected)	0.983	0.524
	0.05 (pooled entropy)	5.729	0.500
TL;DR	1.0 (selected)	0.902	0.528
	0.05 (pooled entropy)	2.650	0.494
SAFERLHF	1.0 (selected)	0.959	0.696
	0.1 (pooled entropy)	3.497	0.617

The two criteria rank the two η_t values identically on all three panels, with worst-objective gaps of 0.024, 0.034 and 0.079. The TL;DR pooled-entropy arm is also the only setting of NBPO we have measured anywhere that falls *below* the reference on one of its objectives. This is not a study of selection rules — three panels and two η_t values each — but it does mean the choice reported in Table 2 is not one the evaluation data was used to make, and is not one the evaluation data contradicts.

What the realization criterion does not see. The criterion of the previous paragraph selects η_t , and it agrees with the evaluation on all three panels there. It does not transfer to the optimiser budget. Doubling the number of updates at the selected η_t , changing nothing else, gives:

panel	held-out nMSE (criterion)		worst W^{eval}	
	300	600	300	600
ULTRAFEEDBACK	0.9833	0.9836	0.524	0.537
TL;DR	0.9022	0.8960	0.528	0.503
SAFERLHF	0.9591	0.9762	0.696	0.688

The two columns agree on SAFERLHF, disagree on TL;DR, and on ULTRAFEEDBACK the criterion cannot separate the two arms at all — its gap is 0.0003 while the evaluation’s is 0.013. One agreement, one disagreement and one abstention is what no information looks like: the held-out realization criterion tells us nothing about how long to train, even though it is informative about η_t . Nor is the budget monotone in the evaluation, which would have been the easy alternative rule: the longer run wins on one panel and loses on the other two.

We therefore keep the same 300-update budget on all three panels and report the budget as an open sensitivity rather than a tuned hyperparameter. That is not a face-saving choice — it costs us the better ULTRAFEEDBACK number — but selecting 600 updates there on the strength of W^{eval} would have been selection on the evaluation set, and the same choice would have cost 0.025 on TL;DR. What separates the two arms is the distance the policy travels ($\|h\|_{\text{rms}}$ 1.89 $\rightarrow$ 2.62 on ULTRAFEEDBACK) rather than how well it realizes the target, and we have no training-side quantity that says how far is far enough.

Does the finite-pool surplus survive on held-out prompts? The dual solve guarantees a positive surplus in every objective *on the frozen pool it is solved on*. Whether that survives on prompts the pool never saw is a separate question, and Algorithm 1’s own acceptance test is the natural way to ask it: $\min_k \hat{s}_k$ measured on a monitoring set, with the monitoring judge, at the same β . We carved 100 such prompts out of each panel’s held-out split — disjoint from the training pairs and from the evaluation prompts — decoded four samples per prompt, and evaluated the regularized game value against the reference.

The answer differs sharply by panel. On SAFERLHF the surpluses are 0.240 and 0.148, comfortably positive, and the gate accepts. On ULTRAFEEDBACK the minimum surplus is -0.006 for the reported

arm and -0.009 for the longer one: the average surplus stays positive ($+0.013$, $+0.015$) but one objective goes negative, Nash welfare is undefined, and Algorithm 1 as written would reject both candidates. The finite-pool guarantee therefore does not transfer uniformly, and the panel where it transfers is the panel where NBPO’s worst-objective win rate is highest (0.696 against 0.524). This is a sharper statement of the limitation the paper already carried in words — that the neural algorithm approximates the population objective from finite samples — and it is a diagnostic a practitioner can run before trusting a fitted arm, since it needs no evaluation data.

Two cautions on reading it. $\hat{s}_k$ and W_k^{eval} are different quantities: the first is a value against the worst regularized opponent at $\beta = 0.25$, judged by Qwen3-32B; the second is a win rate against the reference’s own samples, judged by Llama-3.3-70B. A policy can beat the reference on average, as the reported ULTRAFEEDBACK arm does on all four objectives, while its regularized game-value surplus against an adaptive opponent is slightly negative. And 100 prompts is a small monitoring set: the gap between the two ULTRAFEEDBACK arms, -0.006 against -0.009 , is not resolved at that size, which is the third and last reason we do not use this quantity to select the optimiser budget either.

The pool is not short of disagreement. Table 17 rules out the third explanation. On the frozen pools no response is top under every objective for 46% of ULTRAFEEDBACK prompts, 69% of TL;DR prompts and 23% of SAFERLHF prompts, counting ties as agreement, and the objective-wise orderings disagree on 0.24 to 0.34 of response pairs. Splitting ULTRAFEEDBACK at the median disagreement (Table 18) shows that bargaining is genuinely harder where the objectives conflict — the minimum surplus falls from 0.081 to 0.046 — but that the gap between the adaptive and fixed-reference representations does not widen there.

J JUDGE INSTANTIATION AND PREFERENCE LABELING

Training judges. Use the same prompted pairwise interface on both panels. For each objective, Qwen3-32B compares two responses under an attribute-specific rubric. Each pair is queried in both presentation orders, and the verdicts are swap-averaged. The fixed-reference precursor also contains deterministic Beaver reward and cost heads; those runs should not be conflated with the prompted-model main comparison.

Evaluation judges. Evaluation is separate from training and uses Phi-4 and Llama-3.3-70B. Each evaluator returns $[[A]]$, $[[B]]$, or $[[TIE]]$, scored as 1, 0, or $1/2$, in both presentation orders. This makes the empirical evaluation model skew-symmetric after averaging.

Rubrics. Helpfulness asks which response better accomplishes the user’s stated goal; correctness asks which is more accurate; coherence asks which is clearer and better organized; conciseness asks which conveys the needed content with less padding; and harmlessness asks which is safer. The judge is instructed to evaluate only the designated objective.

Prompt template. For concreteness, the helpfulness system message and pairwise user message are

```
System: You compare two AI responses on HELPFULNESS ONLY:
which better helps the user accomplish their stated goal
(relevance, substance, actionable detail). Ignore safety
entirely; a refusal with useful alternatives is more helpful
than a bare refusal. End with exactly [[A]] or [[B]] or
[[TIE]].
```

```
User: [User prompt] {prompt} [Response A] {A} [Response B]
{B} Verdict:
```

The other objectives use the same template with only the rubric changed, and each pair is queried again after swapping responses A and B.

1350 **From verdicts to regularized opponents.** For a current policy π_t and a reference comparator z ,
1351 estimate

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$$\hat{m}_k^{\pi_t}(z | x) = \frac{1}{L} \sum_{\ell=1}^L (B_k(y_\ell, z) - \frac{1}{2}), \quad y_\ell \sim \pi_t(\cdot | x).$$

1353 Reference responses are reweighted by $\exp\{-\hat{m}_k^{\pi_t}(z | x)/\beta_k\}$ and normalized to sample $z_k \sim \hat{\mathcal{V}}_{k, \pi_t}^*$.
1354 For a policy pair (y, y') sharing z_k , the binary difference in (24) enters the regression target directly.
1355 No scalar reward model or value function is fitted by NBPO.
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Table 8: Data, sampling, evaluation, and NBPO-specific settings.

setting	ULTRAFEEDBACK/Llama	ULTRAFEEDBACK/Qwen TL;DR/Llama	SAFERLHF/Llama
Objective panel	instr. following; truth-helpfulness; truthfulness; coverage; faithfulness; helpfulness; harmless- fulness; honesty; helpful-honesty; instr. following conciseness; helpfulness		
Base/reference policy	Llama-3.1-8B-Instruct	Qwen3-8B	Llama-3.1-8B-Instruct
Training preference judge	Qwen3-32B	Qwen3-32B	Qwen3-32B
Primary evaluation judge	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct
Secondary evaluation judge	Phi-4 (secondary check)	Phi-4 (secondary check)	Phi-4 (secondary check)
Qwen thinking mode	–	reasoning trace suppressed: the chat template emits an empty think block in the generation prompt, equivalent to enable_thinking = False	–
Training prompts	7,000	300	600
Validation split	current: 10% pair-level; final rerun: prompt-level	current: 10% pair-level; final rerun: prompt-level	current: 10% pair-level; final rerun: prompt-level
Held-out evaluation prompts	698 (audited prompt-disjoint)	100	400
Outer proximal stages T	1	1	1
Primary comparison (Table 2); the released arms			
Dual iterations per stage M	4,000	4,000	4,000
Fixed-point iterations per dual step 1	1	1	1
Proximal coefficient η_t	1.0	1.0	1.0
η_t selection rule	lower held-out normalized MSE of (26) on the panel’s own prompt-disjoint pair split (Table 16); it selects 1.0 on all three Llama panels, against 0.05/0.05/0.1 for the pooled-entropy rule. No evaluation data is used by either rule.		
Effective target RMS $\eta_t \ Y\ _{\text{rms}}$	9.69	—	11.49
Regression target	(25) conditional mean	(24) draw	(25)
Pair rows fitted	34,026 (5,671 prompts, after the contamination filter of Appendix E)	37,800	3,132 (522 prompts)
Optimizer steps / learning rate	$300 / 5 \times 10^{-7}$	$300 / 5 \times 10^{-7}$	$300 / 5 \times 10^{-7}$
Batch $\times$ accumulation	2×8	2×8	2×8
Matched ablation (Table 3); Llama panels only			
Fixed-point iterations per dual step 8, damping 0.9	1	8, damping 0.9	8, damping 0.9
Proximal coefficient η_t	0.05	—	0.05
η_t selection rule	largest η_t with pooled entropy $\geq 0.9 \ln I$ (Table 14)	—	same rule
Pair rows fitted	3,204 (534 prompts)	—	3,132 (522 prompts)
Optimizer steps / learning rate	$400 / 1 \times 10^{-7}$	—	$400 / 1 \times 10^{-7}$
Batch $\times$ accumulation	4×4	—	4×4
Control weight normalization	rescaled to none	to none	to 30.5
Dual step size γ_m	0.5	0.05	2.0
Opponent temperatures $\beta_{1:K}$	0.25 (all k)	0.25 (all k)	0.25 (all k)
Dual box $[\lambda_{\min}, \lambda_{\max}]$	$[10^{-3}, 10^3]$	$[10^{-3}, 10^3]$	$[10^{-3}, 10^3]$
Dual initialization/warm start	$\lambda^{(0)} = 1$, no warm start	$\lambda^{(0)} = 1$, no warm start	$\lambda^{(0)} = 1$, no warm start
Dual stopping criterion and tolerance	fixed budget M ; projected KKT residual 5.0×10^{-7} , fixed-point residual 1.7×10^{-10}	fixed budget M ; KKT residual 6.6×10^{-3}	fixed budget M ; KKT fixed point 9.3×10^{-12} , fixed point 5.6×10^{-17}
Policy responses per prompt	4	4	4
Opponent candidate pool per prompt	4	4	4
Reference comparators per prompt	4	4	4
Learner comparisons per comparator L	4	4	4
Response-pair proposal ξ	all 6 unordered pairs	all 6 unordered pairs	all 6 unordered pairs
Presentation orders per judged pair	2	2	2
Judge repetitions per order	1	1	1
Training response temperature	0.9	0.9	0.9
Training response top- p /top- k	0.95 / –	0.95 / –	0.95 / –
Maximum training response tokens	512	512	512
Training-judge temperature	0.0 (greedy)	0.0 (greedy)	0.0 (greedy)
Training-judge top- p /top- k	– / –	– / –	– / –
Maximum judge tokens	512	512	512
Evaluation generation temperature	0.7	0.7	0.7
Evaluation generation top- p /top- k	0.9 / –	0.9 / –	0.9 / –
Maximum evaluation response tokens	1,024	1,024	1,024
IFEval protocol/version	lm-eval-harness 0.4.9, task ifeval (task version 4), 0-shot, chat template applied, greedy decoding; we report prompt-level strict accuracy		
TruthfulQA protocol/version	lm-eval-harness 0.4.9, tasks truthfulqa_mc1 (version 2) and truthfulqa_mc2 (version 3), 0-shot, no chat template; we report MC2 accuracy		
HarmBench protocol/version	not evaluated; harmfulness is scored on SAFERLHF by the same independent pairwise evaluator as every other objective		

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Table 9: Policy optimization, model adaptation, tuning, and compute settings.

setting	ULTRAFEEDBACK/Llama	ULTRAFEEDBACK/Qwen	TL;DR/Llama	SAFERLHF/Llama
Adaptation mode (full/PEFT)	full fine-tuning	full fine-tuning	full fine-tuning	full fine-tuning
Trainable modules	all transformer parameters	all transformer parameters	all transformer parameters	all transformer parameters
Optimizer	Adafactor	Adafactor	Adafactor	Adafactor
Learning rate	5×10^{-7}	5×10^{-7}	5×10^{-7}	5×10^{-7}
Learning-rate scheduler	cosine	cosine	cosine	cosine
Warmup ratio/steps	0.1 / -	0.1 / -	0.1 / -	0.1 / -
Weight decay	0	0	0	0
Per-device batch size	2	2	2	2
Gradient accumulation steps	8	8	8	8
Global batch size	16	16	16	16
Epochs or optimizer updates per stage	300 updates	300 updates	300 updates	300 updates
Maximum input length	1,024	1,024	1,024	1,024
Maximum sequence length	2,048	2,048	2,048	2,048
Gradient clipping	1.0 (framework default)	1.0 (framework default)	1.0 (framework default)	1.0 (framework default)
Numerical precision	bf16	bf16	bf16	bf16
Gradient checkpointing	yes (non-reentrant)	yes (non-reentrant)	yes (non-reentrant)	yes (non-reentrant)
Distributed-training strategy	single device (no sharding)	single device (no sharding)	single device (no sharding)	single device (no sharding)
Random seeds	42	42	42	42
Final independent training seeds	seeds 42, 43, 44 on ULTRAFEEDBACK; worst W^{eval} 0.523 ± 0.005 , avg 0.537 ± 0.005			
Hyperparameters tuned and candidate grid	$\eta_t \in \{0.02, 0.05, 0.1, 0.25\}$ (all panels); learning rate $\in \{5 \times 10^{-9}, 2 \times 10^{-8}, 5 \times 10^{-8}, 1 \times 10^{-7}, 2 \times 10^{-7}, 5 \times 10^{-7}, 2 \times 10^{-6}\}$; optimizer updates $\in \{300, 400, 1200\}$. Reward-model baselines: the interpolation/preference grid of Table 11. Nothing else was swept.			
Validation set and selection metric	η_t : the frozen response pool’s dual solution only, by the pooled-opponent-entropy rule of Table 14; no policy is trained to choose it. Learning rate and updates: the prompt-disjoint <code>pairs.test</code> split of the same pair set, scored by the normalized MSE of (26). Reward-model baselines: 300 held-out validation prompts, largest minimum reference-standardised utility. No selection uses W^{eval} or any evaluation prompt.			
Number of tuning trials	4 dual solves per panel for η_t (no training); 10 trained NBPO fits scored on <code>pairs.test</code> for learning rate and updates (7 on ULTRAFEEDBACK, 3 on SAFERLHF); 7 interpolation points per 4-objective panel and 5 per 2-objective panel for each of Rewarded Soups and Panacea.			
Hardware and number of accelerators	1 × H200 (train); H200, tensor-parallel (judge)	2 × 1 × H200 (train); H200, tensor-parallel (judge)	2 × 1 × H200 (train); H200, tensor-parallel (judge)	2 × 1 × H200 (train); H200, tensor-parallel (judge)
Wall-clock time/GPU-hours per run	50 min (0.83 H200-h)	50 min (0.83 H200-h)	37 min (0.62 H200-h)	37 min (0.62 H200-h)

Table 10: Overlap between each arm’s training prompts and its panel’s evaluation prompts, by prompt-text hash. Lower is better; zero is a clean prompt-level split.

panel (evaluation prompts)	arm group	overlap
ULTRAFEEDBACK (698)	released NBPO, before the rebuild	629
	rebuild NBPO (reported)	0
	matched ablation arms (Table 3)	51
	fixed-reference and adaptive-game controls	0
	MaxMin-RLHF, Rewarded Soups, Panacea	0
TL;DR (400)	every arm	0
SAFERLHF (400)	every arm	≤ 1
ULTRAFEEDBACK/Qwen (100)	every arm	99

Table 11: Implementation record for the reward-model-based multi-objective baselines. All three share the preference arms’ response pool, initialisation and 300-update budget, and differ only in how the objective-wise rewards are turned into one policy.

method	reward model(s) and training data	policy optimizer and KL coefficient	trade-off/selection hyper-parameters	selected operating point and compute
MaxMin-RLHF	ArmoRM-Llama3-8B-v0.1 objective heads on the shared response pool; the panel’s objectives are the known groups, so Algorithm 2’s EM over latent user clusters is not needed	DPO on the worst objective’s reward-ranked pairs, $\beta = 0.1$, reference fixed at π_{ref} , Adafactor, lr 5×10^{-7}	adversarial mixture/step size: hard arg min over reference-standardised improvements, $T = 3$ rounds of 100 updates; at $t = 0$ every improvement is zero and the tie is broken on the declared objective order (raw levels cannot break it: one panel’s conciseness head is a negated verbosity score)	selected objectives UF/Llama instr./honesty/truth., UF/Qwen instr./truth./truth., TL;DR cover./concise/help., SAFERLHF help./harm/help.; 2×300 validation samples per round; ≈ 1.3 H200-hours per panel
Rewarded Soups	same reward heads; one expert per objective, all from the panel’s own shared initialisation (Llama-3.1-8B-Instruct, or Qwen3-8B on the Qwen panel)	DPO per expert, $\beta = 0.1$, 300 updates each, then linear weight interpolation	interpolation grid and validation rule: uniform, K one-heavy (0.55) and 2 Dirichlet points; largest minimum reference-standardised validation utility	$\lambda = (.55, .15, .15, .15)$ UF/Llama, (.15, .15, .15, .55) UF/Qwen, TL;DR, (.5, .5) SAFERLHF; 298/100/100/100 held-out validation prompts; ≈ 3.3 H200-hours per 4-objective panel
Panacea	same reward heads; the per-objective DPO objectives are aggregated online rather than pre-mixed	SVD-LoRA, $k = 8$ preference-agnostic singular values, $m = K$ injected as the remaining singular values with a learnable per-matrix scale, AdaLoRA orthogonality penalty, LS aggregation, $\beta = 0.1$, lr 2×10^{-4} , 300 updates	preference-conditioning grid and validation rule: same grid as Rewarded Soups; simplex vertices upweighted during training; largest minimum reference-standardised validation utility	$\lambda = (.15, .15, .55, .15)$ UF/Llama, (.15, .55, .15, .15) UF/Qwen, (.15, .15, .55, .15) TL;DR, (.4, .6) SAFERLHF; ≈ 1.2 H200-hours per panel

Table 12: Safety transfer on HarmBench: official attack-success rate (ASR; lower is better) over the 300 standard and contextual behaviours, generated greedily and scored by the official HarmBench-Llama-2-13b-cl s classifier, using the same frozen checkpoints as the main comparison.

method	HarmBench ASR ↓
Base = $\pi_{\text{ref}} = \pi_0$	0.223
NBPO (ours)	0.290
PROSPER/MaxEntBW	0.310
Utilitarian	0.350
Absolute maxmin	0.433
Surplus maxmin	0.320
Uniform-INPO	0.377
MOPO	0.240
MaxMin-RLHF	0.463
Rewarded Soups	0.453
Panacea	0.537

Table 13: Optimization sensitivity of the rebuilt ULTRAFEEDBACK solve at $\eta_t = 0.05$. Every row is a full dual solve on the same frozen tensor and differs in one setting. Opponent entropy is measured over the four comparators ($\ln 4 = 1.386$ is uniform), the fixed-point residual measures policy-opponent consistency in (21), and the dual residual is $\|\hat{s} - \lambda^{-1}\|_\infty$. Two things are worth reading off. The opponent budget R does not change the solution at this step size — it only sharpens the fixed-point residual — so the residual of 0.9999 reported for $R = 1$ elsewhere in this paper is a consequence of the degenerate step size at $\eta_t = 1$, not an independent defect. And the surplus rises monotonically as the opponent is weakened, reaching its maximum when $\beta \rightarrow \infty$ makes the opponent exactly uniform: surpluses measured against different opponents are not comparable to each other.

ablation	setting	$\min_k s_k$	$\text{avg. } s_k$	$\sum_k \log s_k$	opp. entropy	fixed-point res.	dual res.
Opponent temperature β <i>default</i>	$\beta = 0.15$	0.06572	0.07302	-10.4796	0.963	1.8×10^{-10}	4.6×10^{-7}
	$\beta = 0.25$	0.06579	0.07297	-10.4823	1.160	1.7×10^{-10}	5.0×10^{-7}
	$\beta = 0.50$	0.06654	0.07360	-10.4473	1.314	1.4×10^{-10}	4.2×10^{-7}
	$\beta = 1.00$	0.06719	0.07423	-10.4134	1.367	1.2×10^{-10}	3.6×10^{-7}
	$\beta \rightarrow \infty$ (fixed reference)	0.06790	0.07494	-10.3749	1.386	1.0×10^{-10}	2.9×10^{-7}
Fixed-point budget R <i>default</i>	$R = 1$, no damping	0.06579	0.07297	-10.4823	1.160	2.0×10^{-9}	5.0×10^{-7}
	$R = 8$, no damping	0.06579	0.07297	-10.4823	1.160	1.7×10^{-10}	5.0×10^{-7}
	$R = 8$, damping 0.9	0.06579	0.07297	-10.4823	1.160	1.7×10^{-10}	5.0×10^{-7}
	$R = 64$, damping 0.9	0.06579	0.07297	-10.4823	1.160	2.1×10^{-13}	5.0×10^{-7}
Dual budget M <i>default</i>	$M = 400$	0.06163	0.06883	-10.7176	1.160	1.3×10^{-5}	1.3×10^{-2}
	$M = 4,000$	0.06579	0.07297	-10.4823	1.160	1.7×10^{-10}	5.0×10^{-7}
	$M = 40,000$	0.06579	0.07297	-10.4823	1.160	1.1×10^{-16}	1.8×10^{-15}

Table 14: Selecting the proximal step size η_t on a training-side criterion. For each candidate the dual of (27) is solved to the opponent fixed point on that panel’s frozen tensor, and the concentration of the resulting exact inner solution is read off. The rule is fixed in advance: take the largest η_t whose pooled policy keeps at least 0.9 of the uniform entropy, so that the $(1/\eta_t)$ trust region still binds. No evaluation prompt is decoded. The released $\eta_t = 1$ is off this table entirely: at $\eta_t = 1$ the product $\eta_t \sum_k \lambda_k$ is 25 on a four-response pool and the inner solution is a point mass, so the regression target of (26) is built from a degenerate pooled policy.

panel	η_t	$\eta_t \sum_k \lambda_k$	entropy / $\ln I$	$\max_i \pi_i^*$	$\min_k s_k$	fixed-point res.
ULTRAFEEDBACK	0.02	1.67	0.972	0.612	0.0434	2.4e-8
	0.05	2.76	0.933	0.810	0.0658	1.7e-10
	0.1	4.14	0.873	0.938	0.0873	2.4e-13
	0.25	7.61	0.732	0.997	0.1176	1.1e-16
TL;DR	0.02	1.95	0.972	0.537	0.0290	5.3e-10
	0.05	3.23	0.933	0.698	0.0433	1.3e-14
	0.1	4.86	0.874	0.844	0.0569	5.6e-17
	0.25	8.98	0.734	0.976	0.0757	1.1e-16
SAFERLHF	0.02	1.29	0.986	0.518	0.0302	1.5e-10
	0.05	2.08	0.965	0.671	0.0468	8.9e-16
	0.1	3.05	0.933	0.810	0.0639	5.6e-17
	0.25	5.33	0.850	0.958	0.0911	5.6e-17

Table 15: Exact finite-pool solutions under each aggregation rule, evaluated with all three welfare functionals. Every solve shares the panel’s tensor, β , disagreement point, response pool, η_t , dual budget and opponent fixed point ($R = 8$, damping 0.9; residuals below 10^{-9} throughout). The Nash dual carries raw λ with $\|\lambda\|_1 = \sum_k 1/s_k$, while the utilitarian rule sits at $\lambda = 1$ and the max-min rules on the simplex, so the control rules are rescaled to the Nash arm’s $\|\lambda\|_1$ on the same panel: without that the rules take proximal steps differing by a factor of 14 to 55 and the comparison is about step size rather than about the direction of the weight vector. Nash itself is never rescaled — its magnitude is fixed by $\lambda_k = 1/s_k$. The fixed-reference row is $\beta \rightarrow \infty$, whose surplus is measured against a collapsed opponent and is therefore not comparable to the rows above it.

panel	aggregation rule	$\min_k s_k$	avg. s_k	$\sum_k \log s_k$	$\lambda_{\max}/\lambda_{\min}$
ULTRAFEEDBACK	Nash	0.06579	0.07297	-10.4823	1.22
	utilitarian	0.06510	0.07340	-10.4631	1.00
	absolute max-min	0.06311	0.07485	-10.4066	2.39
	surplus max-min	0.06805	0.07153	-10.5532	2.29
	<i>fixed reference</i>	0.06790	0.07494	-10.3749	1.21
TL;DR	Nash	0.04334	0.06556	-11.0037	1.83
	utilitarian	0.03861	0.06997	-10.8440	1.00
	absolute max-min	0.03689	0.06989	-10.8451	2.40
	surplus max-min	0.05129	0.05678	-11.4822	5.47
	<i>fixed reference</i>	0.04562	0.06858	-10.8198	1.82
SAFERLHF	Nash	0.06387	0.06560	-5.4490	1.05
	utilitarian	0.06347	0.06566	-5.4477	1.00
	absolute max-min	0.05949	0.06605	-5.4446	1.68
	surplus max-min	0.06484	0.06544	-5.4532	1.20
	<i>fixed reference</i>	0.06905	0.07060	-5.3020	1.04

Table 16: Regression-realization diagnostics for the pairwise target in (26), on the training pairs and on a prompt-disjoint held-out split of the same pair set. The zero-predictor MSE is $\mathbb{E}[Y^2]$ for $Y = \eta_t \sum_k \lambda_k Z_{t,k}$, attained by $\pi = \pi_t$, so a normalized MSE below 1 means the fitted policy beats doing nothing. η_t sets the size of that target: the target RMS column is $\eta_t \|Y\|_{\text{rms}}$. Sign agreement is reported on the pairs whose target is non-zero. The last row is a control in which μ is precomputed twice and the two passes differenced, which bounds the measurement error of every entry above it. Nothing here uses evaluation data: the selected η_t is the one with the lower held-out normalized MSE on its own panel, and it is 1.0 on all three.

panel	η_t		target RMS	$\ h\ _{\text{rms}}$	training pairs		held-out pairs		corr.	sign agr.
					nMSE	R^2	nMSE	R^2		
ULTRAFEEDBACK	1.0	<i>contaminated</i>	9.74	2.79	1.0005	-0.0005	0.9977	+0.0023	0.148	0.558
ULTRAFEEDBACK	1.0	<i>selected, rebuilt</i>	9.69	1.89	0.9295	+0.0705	0.9833	+0.0167	0.140	0.445
	0.05	<i>pooled entropy</i>	0.48	1.04	46.7943	-45.7943	5.7286	-4.7286	-0.025	0.404
TL;DR	1.0	<i>selected</i>	11.49	2.85	0.5189	+0.4811	0.9022	+0.0978	0.327	0.553
	0.05	<i>pooled entropy</i>	0.57	0.69	2.4293	-1.4293	2.6497	-1.6497	-0.082	0.400
SAFERLHF	1.0	<i>selected</i>	6.45	2.74	0.7069	+0.2931	0.9591	+0.0409	0.272	0.305
	0.1	<i>pooled entropy</i>	0.65	1.05	8.1346	-7.1346	3.4974	-2.4974	0.047	0.303
Control: μ twice	—	<i>floor</i>	—	0.00	1.0000	0.0000	1.0000	0.0000	—	—

Table 17: Disagreement between objectives on the frozen response pools. For each prompt, every learner sample is scored by its mean margin against the reference pool under each objective; the weak no-Condorcet rate is the share of prompts on which no sample is top under *every* objective, with ties counted as agreement, and ranking disagreement is the mean pairwise Kendall-tau distance between the objective-wise orderings.

panel	K	prompts	weak no-Condorcet	ranking disagreement
ULTRAFEEDBACK	4	7000	0.461	0.236
TL;DR	4	582	0.686	0.339
SAFERLHF	2	600	0.228	0.311

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Table 18: ULTRAFEEDBACK prompts split at the median ranking disagreement, with the adaptive game ($\beta=0.25$) and fixed-reference ($\beta \rightarrow \infty$) duals solved on each half to the same opponent fixed point. Bargaining is harder where the objectives disagree, but the surplus gap between the two representations does not widen there. The two surpluses are *not* comparable to each other: each is measured against its own opponent, and $\beta \rightarrow \infty$ collapses that opponent onto π_{ref} .

prompt subset	prompts	adaptive min s	fixed-ref. min s	difference
Low disagreement	3597	0.0815	0.0838	-0.0023
High disagreement	3403	0.0459	0.0477	-0.0018

Table 19: Existing NBPO-aligned component diagnostic on 698 ULTRAFEEDBACK prompts. These game-surplus values are secondary to the independent metrics in the main text.

variant	min s	avg. s	$F(\pi)$	avg. ref. win
NBPO	0.028	0.056	-11.82	0.556
Fixed-reference NBPO	0.040	0.061	-11.40	0.561
Game-utilitarian	-0.073	-0.048	-	0.452
Game-surplus-maxmin	-0.044	-0.006	-	0.494